

Jimmy Pan

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Education

University of British Columbia

Expected May 2029

BASc – Engineering Physics (GPA: 4.22)

Awards: Trek Excellence Scholarship (2025), Dean's Scholar (2025), Dean's List (2026)

Coursework: Instrument Design (Robotics), Computing Systems, Software Construction, Data Structures and Algorithms, Signals and Systems, Probability with Physical Applications, Applied Linear Algebra

Experience

UBC Rocket

Jul 2025 – Present

Firmware Co-Lead

Vancouver, BC

- Developed and refactored an **STM32 servo driver** using STM32 HAL, hardware timers, and PWM to implement servo-control functionality and improve code maintainability.
- Refactored embedded firmware modules, standardized APIs, and developed reusable mathematical and utility functions; resolved bugs and improved testing and debugging infrastructure.
- Contributed to **DShot ESC protocol** implementation by developing packet-generation code, testing with ESC and motor hardware, and assisting with firmware integration.
- Developed **C++/Qt Quick (QML)** Ground Control Station software, implementing application UI and telemetry logging; tested and debugged functionality, and packaged the application and dependencies for Linux deployment.
- Led firmware team recruitment, including reviewing applications, conducting technical interviews, and coordinating candidate outreach.

Projects

Crabby – Autonomous Competition Robot | C/C++, ESP32-S3, Embedded Systems, PCB Design, Robotics

May – Aug 2026

- Led **firmware development and architecture** for an autonomous competition robot using two **ESP32-S3** microcontrollers and a **Raspberry Pi**; team placed **3rd overall**, and the robot demonstrated all five competition tasks.
- Developed **PWM/DIR/BRAKE motor control** and encoder, servo, and tape-sensor drivers; implemented **I²C communication** for ToF LiDARs and integrated stepper-motor control.
- Designed and implemented a framed **UART protocol** connecting two ESP32-S3s and a Raspberry Pi, with packet parsing, command handling, and error recovery; developed **non-blocking control architecture** for coordinated autonomous execution.
- Integrated and tested dual **PMW3610 optical sensors** for relative position and heading estimation, implementing SPI communication and supporting encoder-based motion control and environmental alignment.
- Designed, assembled, and validated **four custom PCBs** for power distribution, motor control, H-bridge drivers, and tape sensing; integrated all four boards into the competition robot.
- Integrated **Raspberry Pi vision, motor-control firmware, and mechanical subsystems**; performed full-system testing and debugged hardware–software interactions during competition preparation.

Closed-Loop RPM Motor Control System | Digital Logic, Analog Circuits, PI Control, Circuit Debugging

Nov – Dec 2025

- Built and tested a **closed-loop motor-speed controller** integrating a phototransistor sensor, 8-bit counter, R-2R DAC, analog PI controller, and BJT motor driver.
- Assembled and debugged **latch/reset timing and frequency-to-voltage measurement circuitry**; verified counter, DAC, and sampling behavior using Digilent AD3 instrumentation.
- Validated closed-loop speed regulation across motor-speed setpoints and mechanical load changes; measured RPM using optical pulse counts and characterized the controller's response.

Weighted Graph Library | Java, Data Structures, Algorithms, JUnit

Oct 2025

- Implemented **adjacency-list and adjacency-matrix** graph representations in Java, including algorithms for minimum-cost paths, graph diameter, and graph center.
- Developed **JUnit test suites** covering graph operations, edge cases, exceptions, and path algorithms across both implementations, achieving **96.46% overall test coverage and 97.33% branch coverage**.

Skills

Programming & Software: C/C++, Java, Python, QML, MATLAB, Git/GitHub, CMake, Linux

Embedded & Hardware: ESP32-S3, STM32, Raspberry Pi, UART, SPI, I²C, PWM, Device Drivers, PCB Design, Motor Control, Analog/Digital Circuits

Tools & Testing: PlatformIO, ESP-IDF, STM32CubeIDE, STM32CubeMX, Altium, KiCad, Quartus, Qt Creator, Digilent WaveForms, Oscilloscopes, Logic Analyzers, JUnit